

Manufacturing & Video-Analytics Intelligent Report

“Generate a comprehensive Manufacturing Production Performance Report using MES, Video Analytics, and InfluxDB data. Include Work Order status with planned vs. completed quantities, machine-wise utilization from CapacityAnalysis, shift-wise production from WeighingMachine, material-wise scrap from Scrap, rejection reasons from Rejected, and live machine health from InfluxDB ECE2 electrical_params (voltage, current, temperature, vibration, and cumulative cycles) and mps measurements (total count, good count, reject count, cycle count, run status, and ideal cycle time). Provide summaries, charts, key insights, machine performance analysis, and actionable recommendations”

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Time filter applied

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Executive Summary



Matched keywords: capacity, count, cumulative, current, cycle, cycles, downtime, ece2, electrical_params, good, ideal, machine, material, mps, production, reject, run, shift, status, temperature, time, total, utilization, vibration, voltage, work_order, worker, workorder

Telemetry window: 2026-09-24T08:50:03Z → 2026-09-25T08:50:03Z (bucketed every 5m)

Work Order Summary

The WorkOrder table shows two status categories. Completed orders total 1,000 units, while Planned orders total 500 units. This indicates that 66.7% of the planned production has already been completed. No other statuses are present, suggesting all active work orders are either finished or still in planning. The ratio highlights good progress but also a backlog of 500 units yet to start.

Data preview — WorkOrder (showing 2 of 2 points/rows)

Status	sum_PlannedQty	record_count
Completed	1000.0000	1
Planned	500.0000	1

Machine Utilization

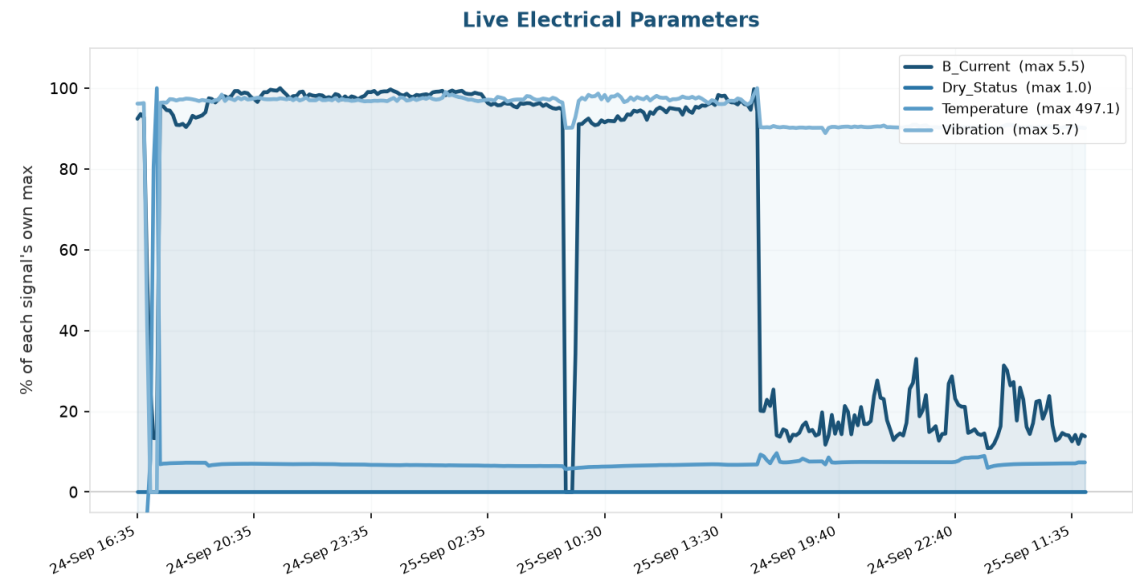
CapacityAnalysis aggregates capacity usage per machine over the queried period. Machine 6 contributed the highest summed CapacityId of 69, followed by Machine 1 with 62, Machine 3 with 41, Machine 2 with 39, and Machine 8 with 20. This distribution highlights that machines 6 and 1 are the most utilized, while Machine 8 shows relatively low usage, possibly indicating underutilization or downtime.

Data preview — CapacityAnalysis (showing 5 of 5 points/rows)

MachinelId	sum_CapacityId	record_count
6	69	3
1	62	3
3	41	2
2	39	2
8	20	1

Live Electrical Parameters

Electrical telemetry from measurement 'electrical_params' contains 293 points over the last ~21 h. Current (B_Current) ranged from 0 to 5.515 A with an average of 3.703 A and a falling trend toward the last value of 0.761 A, suggesting the machine is winding down. Temperature varied widely from -126.6 °C to 497.1 °C, averaging 35.65 °C and currently rising (last 36.48 °C). Vibration stayed between 0 and 5.702 mm/s, averaging 5.337 mm/s and decreasing toward the last 5.139 mm/s. Dry_Status remained constantly 0, indicating the dryer is off. The rising temperature combined with decreasing vibration may signal heating while the machine slows, warranting monitoring.



Telemetry preview — electrical_params (showing 10 of 293 points/rows)

_time	device_id	B_Current	Dry_Status	Temperature	Vibration
2026-09-24 11:05:00+00:00	00WH9OUI	5.097014925373132	0.0	-123.12985074626867	5.482720859015166
2026-09-24 11:10:00+00:00	00WH9OUI	5.162000000000004	0.0	-120.10533333333334	5.483856337865194
2026-09-24 11:15:00+00:00	00WH9OUI	5.115333333333331	0.0	-126.57466666666677	5.491382967631022
2026-09-24 11:20:00+00:00	00WH9OUI	2.9991452991453014	0.0	-28.394871794871847	2.351515965583997
2026-09-24 11:25:00+00:00	00WH9OUI	1.4186666666666692	0.0	39.261333333333337	0.0
2026-09-24 11:30:00+00:00	00WH9OUI	0.7366666666666682	0.0	400.29733333333333	0.0
2026-09-24 11:35:00+00:00	00WH9OUI	0.7358974358974367	0.0	497.1324786324784	0.0
2026-09-24 11:40:00+00:00	00WH9OUI	5.281609195402305	0.0	34.235632183908066	5.49489330971378
2026-09-24 11:45:00+00:00	00WH9OUI	5.266666666666664	0.0	34.955333333333336	5.49951594988505
2026-09-24 11:50:00+00:00	00WH9OUI	5.216000000000011	0.0	35.123999999999997	5.496886746088664

Shift-wise Production (WeighingMachine)

Shift-wise production data from the WeighingMachine table is not available in the current query results. This gap could be due to missing data collection, misnamed table, or a filter that excluded records. Verify the data pipeline and ensure the weighing stations are reporting.

Material-wise Scrap

Material-wise scrap information from the Scrap table is absent. Without scrap quantities per material, it is impossible to assess waste patterns. Check the scrap logging process and confirm the table is populated.

Rejection Reasons

Rejection reasons from the Rejected table were not returned. Lack of rejection data prevents root cause analysis of quality issues. Ensure the rejection capture system is active and data is being stored.

MPS Telemetry

Telemetry for the mps measurement (total count, good count, reject count, cycle count, run status, ideal cycle time) returned no rows for any time window. This suggests a configuration or ingestion problem for the mps measurement. Investigate the data source and InfluxDB write path.

Appendix — Data Quality

Table	Records	Completeness	Outliers	Score
attendances	1	100.0%	0	Good
employees	7	100.0%	0	Good
CapacityAnalysis	5	100.0%	0	Good
MachineMaster	25	89.3%	0	Fair
WorkOrder	2	100.0%	0	Good
WorkOrderLog	1	75.0%	0	Needs review
MachineMaterialMapping	8	100.0%	0	Good
Machine	10	90.0%	0	Fair
MachineFGMapping	39	68.6%	0	Needs review

LLM Usage & Cost

Model	Input tokens	Output tokens	Total tokens	Estimated cost
groq/openai/gpt-oss-120b	4,769	2,267	7,036	N/A

Aggregation Fallbacks

[mes.Machine] Aggregate query failed (('42000', '[42000] [Microsoft][ODBC Driver 18 for SQL Server][SQL Server]Operand data type nvarchar is invalid for sum operator. (8117) (SQLExecDirectW)')) - served raw rows instead.

InfluxDB Measurements With No Data

[influxdb.BM-01] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-02] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-03] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-04] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-05] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-06] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-07] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-08] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-09] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-10] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-11] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows